Operational Properties & Joint Distributions

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OPERATIONAL PROPERTIES & JOINT DISTRIBUTIONS NOTES
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1) OPERATIONAL PROPERTIES OF RANDOM VARIABLES
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Expectation of Sums:

E[X + Y] = E[X] + E[Y]

Important Points:
- Does NOT require independence.
- Works for both discrete and continuous random variables.
- Derived using joint distribution:

E[X+Y] = Sum_x Sum_y (x + y)p(x,y)

This property is called LINEARITY OF EXPECTATION.


Variance of a Scaled Variable:

Variance definition:
Var(X) = E[X^2] - (E[X])^2

Scaling rule:
Var(aX) = a^2 Var(X)

Why?

Var(aX) = E[(aX - a mu)^2]
        = E[a^2 (X - mu)^2]
        = a^2 Var(X)

Key Concept:
Variance measures spread. Scaling by 'a' stretches spread by a^2.


2) JOINT PROBABILITY DISTRIBUTIONS
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Joint distribution models co-occurrence of two random variables.

Discrete Case:
p(x,y) = P(X = x, Y = y)

Total probability condition:
Sum_x Sum_y p(x,y) = 1


Continuous Case:
Double integral of f(x,y) dx dy = 1


3) MARGINAL DISTRIBUTIONS
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Marginal distribution means ignoring the other variable.

Discrete Case:
p(X = x) = Sum_y p(x,y)

Row sums give marginal of X.
Column sums give marginal of Y.


Continuous Case:
f_X(x) = Integral f(x,y) dy
f_Y(y) = Integral f(x,y) dx

This is called integrating out the other variable.
4) CONDITIONAL PROBABILITY / DENSITY
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Definition:
$f(y \mid x) = f(x,y) / f(x)$, provided $f(x)$ is not zero.

Interpretation:
It is like taking a slice of the joint distribution.

Example form:
$f(y|x) = (10 \times y^2) / [(10/3) \times (1 - x^3)]$

5) CONDITIONAL EXPECTATION

Definition:
$E[X \mid Y]$

Meaning:
Expected value of X when Y is fixed.

Law of Iterated Expectation:
$E[X] = E[\, E[X \mid Y] \,]$

Connected concepts:
- LOTUS (Law of the Unconscious Statistician)

6) COVARIANCE & UNCORRELATED VARIABLES

Covariance Definition:
$Cov(X,Y) = E[(X - E[X])(Y - E[Y])]$

Equivalent form:
$Cov(X,Y) = E[XY] - E[X]E[Y]$

Uncorrelated condition:
If $Cov(X,Y) = 0$, then X and Y are uncorrelated.

If X and Y are independent:
$E[XY] = E[X]E[Y]$
Therefore $Cov(X,Y) = 0$.